Cache la Poudre River Metabolism

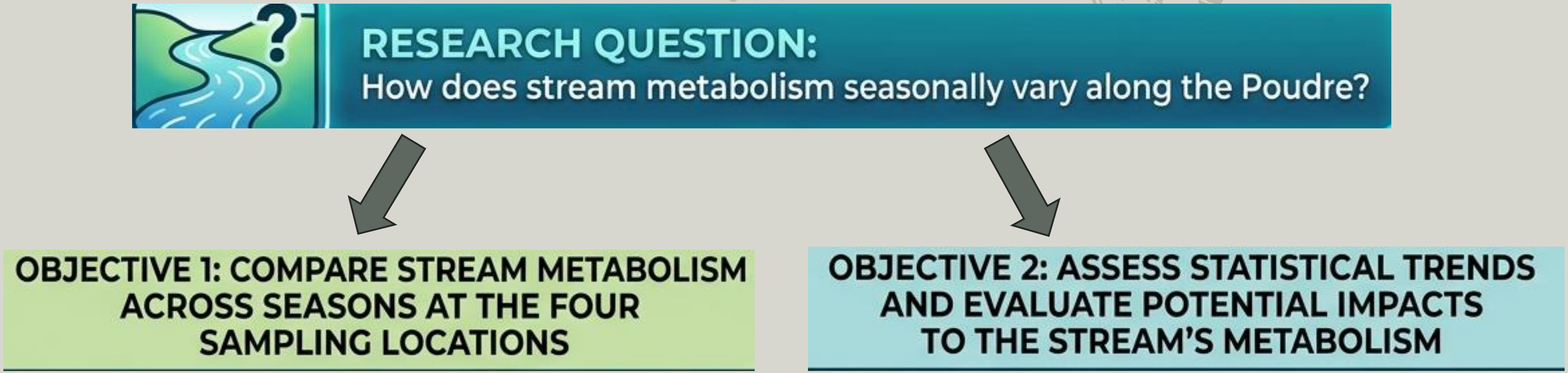
Maggie Rouin

Why Metabolism?

- Beneficial uses of streams:
  - Drinking water
  - Infrastructure
  - Recreation
  - Ecosystems for aquatic life
- Cache la Poudre serves over 400,000
- Stream metabolism indicates overall stream ecosystem health

Source: Poudre Heritage

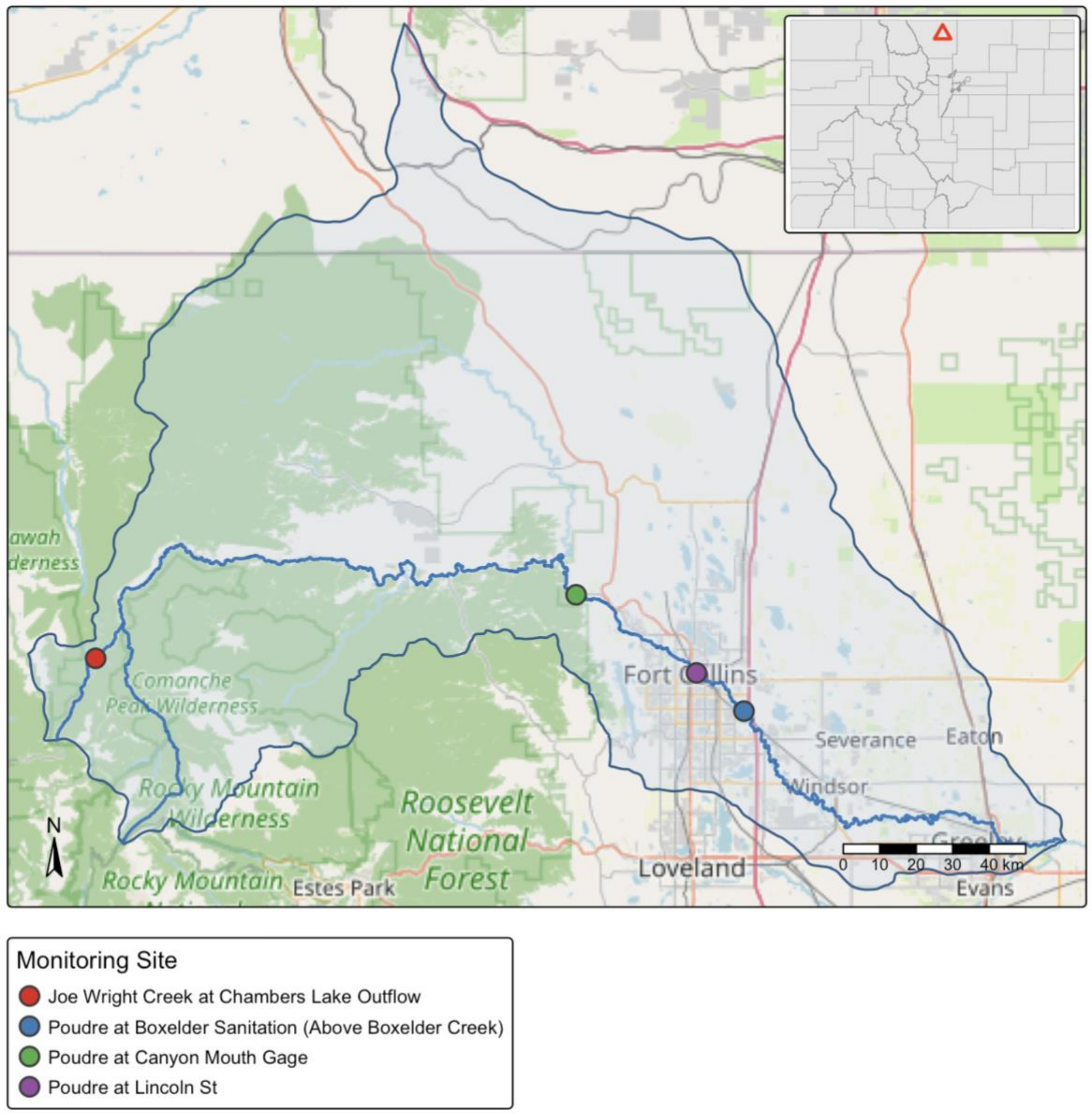
Objectives



Source: Gemini AI

Site Description

- Elevation: 4,870 - 9,150 ft
- Drainage area: 3,000 square miles
- Sampling points: Chambers Lake -> Environmental Learning Center
- Variety of land use throughout



Background

- Stream metabolism
- GPP: gross primary production
- ER: ecosystem respiration
- NEP: net ecosystem production
  - NEP = GPP - ER

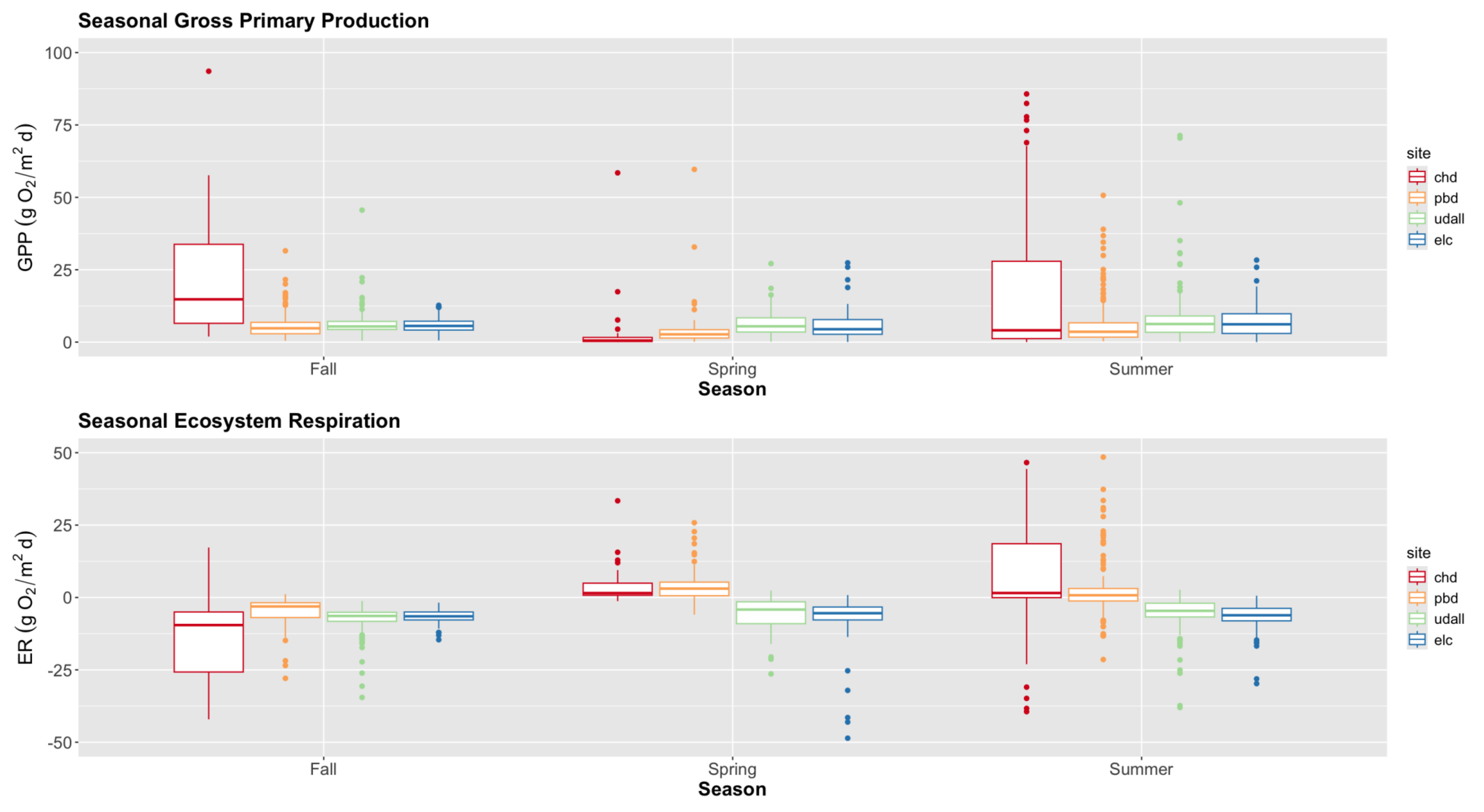
Source: Poudre Heritage

Methods

- Data
  - ROSSyndicate 15-minute continuous DO and temperature
- Model
  - USGS streamMetabolizer
  - Calculations:
    - Solar time
    - Air pressure
    - Turbidity constants
- Statistical analysis
  - Kruskal-Wallis
  - Dunn test

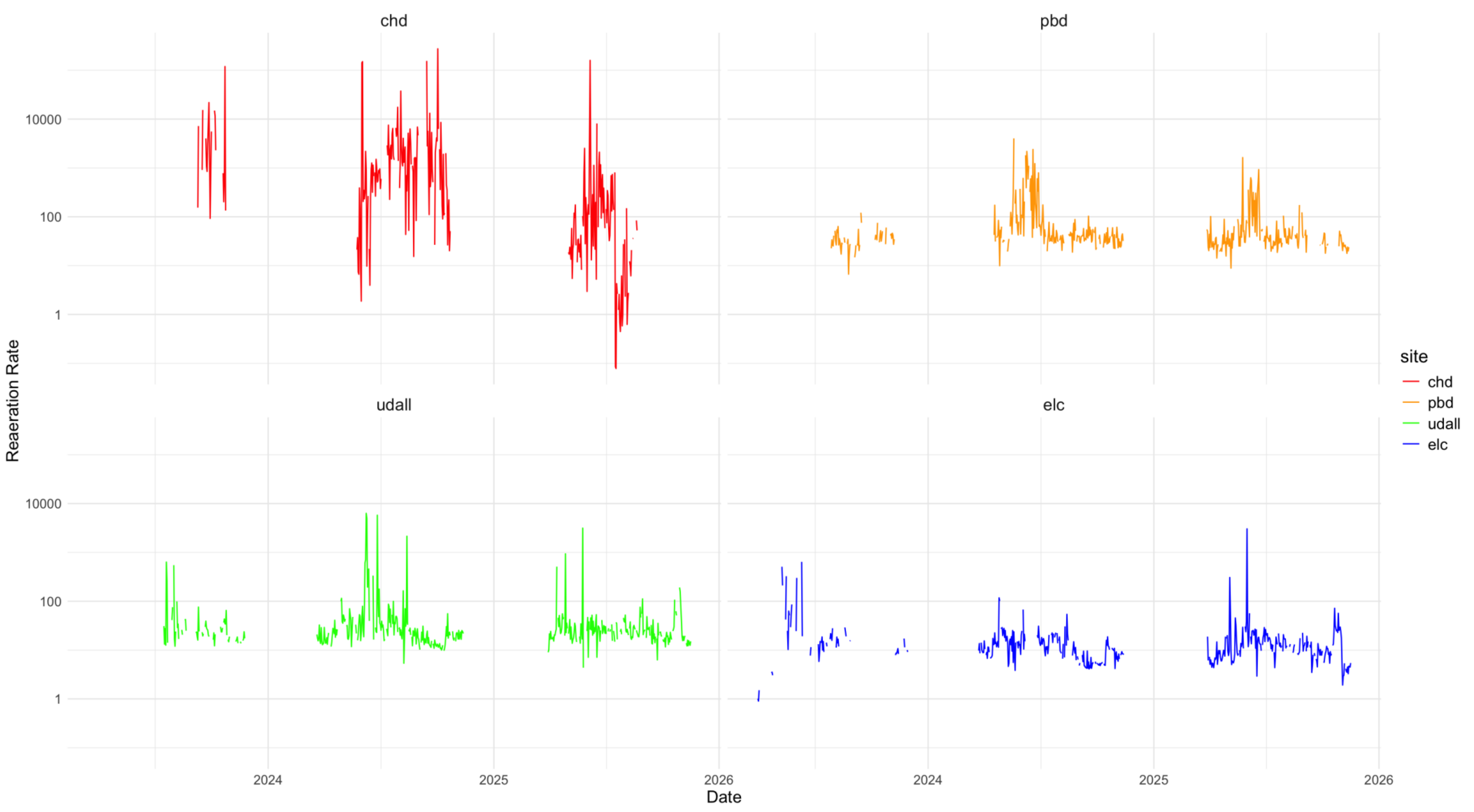
DOI-USGS / streamMetabolizer

GPP vs ER

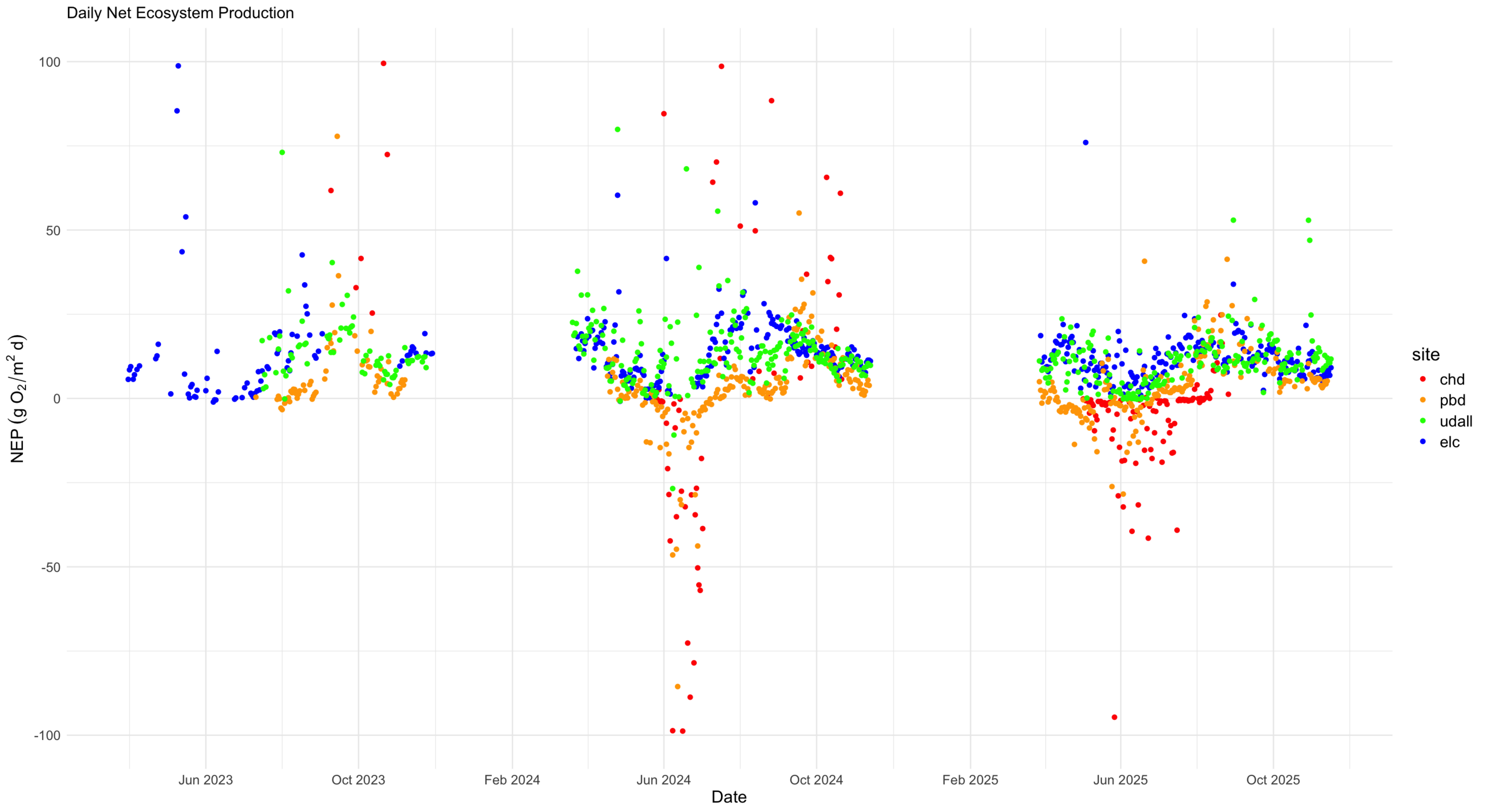


Seasonal Gross Primary Production
GPP (g O_2/m^2 d)
100
75
50
25
0
Fall
Spring
Summer
Season
site
chd
pbd
udall
elc
Seasonal Ecosystem Respiration
ER (g O_2/m^2 d)
50
25
0
-25
-50
Fall
Spring
Summer
Season
site
chd
pbd
udall
elc

Outlier Investigation



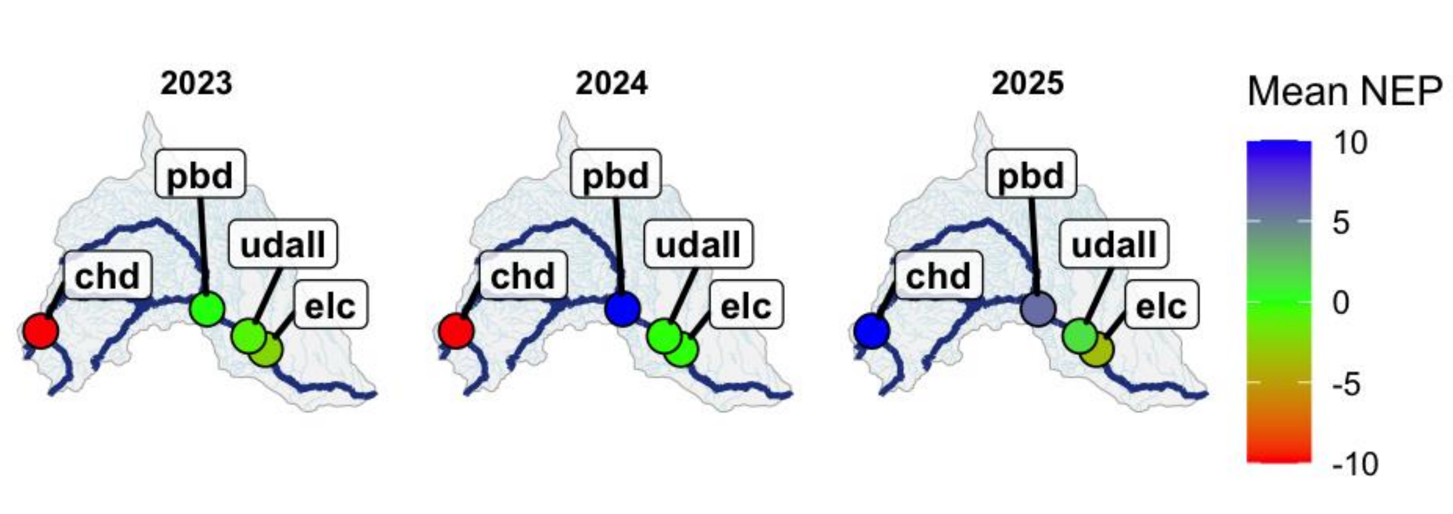
NEP



Daily Net Ecosystem Production
NEP (g O_2/m^2 d)
100
50
0
-50
-100
Jun 2023
Oct 2023
Feb 2024
Jun 2024
Oct 2024
Feb 2025
Jun 2025
Oct 2025
Date
site
chd
pbd
udall
elc

Conclusion

- Key findings
  - GPP was highest at CHD
  - ER was highest at PBD
  - NEP was variable across years
- Limitations
  - Limited data years
  - Partial year view
  - Variability across sites
- Paths forward
  - More data years
  - Different region for the study
  - Review nutrient data



Questions

Sources

- Beaulieu, J. J., Arango, C. P., Balz, D. A., & Shuster, W. D. (2013). Continuous monitoring reveals multiple controls on ecosystem metabolism in a suburban stream. Freshwater Biology, 58(5), 918–937.
- Mulholland, P. J., Fellows, C. S., Tank, J. L., Grimm, N. B., Webster, J. R., Hamilton, S. K., Martí, E., Ashkenas, L., Bowden, W. B., Dodds, W. K., Mcdowell, W. H., Paul, M. J., & Peterson, B. J. (2001). Inter-biome comparison of factors controlling stream metabolism. Freshwater Biology, 46(11), 1503–1517.
- Reed, A. P., Stets, E. G., Murphy, S. F., & Mullins, E. A. (2021). Aquatic-terrestrial linkages control metabolism and carbon dynamics in a mid-sized, urban stream influenced by snowmelt. Journal of Geophysical Research: Biogeosciences, 126, e2021JG006296.
- Winslow, L. A., Zwart, J. A., Batt, R. D., Dugan, H. A., Woolway, R. I., Corman, J. R., … Read, J. S. (2016). LakeMetabolizerWoolway, R. I., Corman, J. R., … Read, J. S. (2016). LakeMetabolizer: an R package for estimating lake metabolism from free- water oxygen using diverse statistical models. *Inland Waters*, *6*(4), 622–636.